

Erdős Problem #830

EP-830

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

We say that $a, b \in \mathbb{N}$ are an amicable pair if $\sigma(a) = \sigma(b) = a + b$. Are there infinitely many amicable pairs? If $A(x)$ counts the number of amicable $1 \leq a \leq b \leq x$ then is it true that $A(x) > x^{1-o(1)}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>